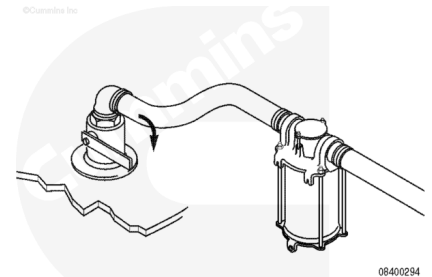


Trainings ▾

Preparatory Steps

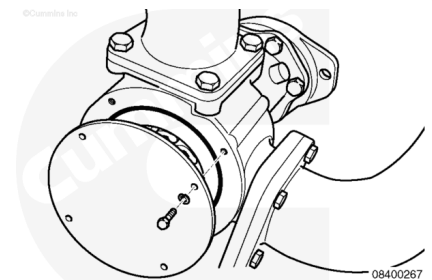
Shut off the sea water inlet valve.



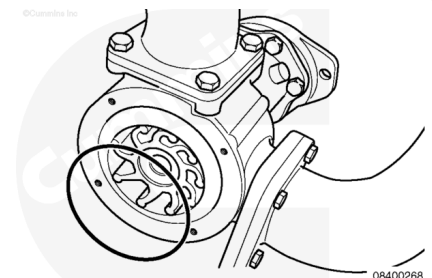
Remove

Jabsco®

Remove the four capscrews with lock washers, as well as the cover, from the sea water pump. Use a 10-mm socket.



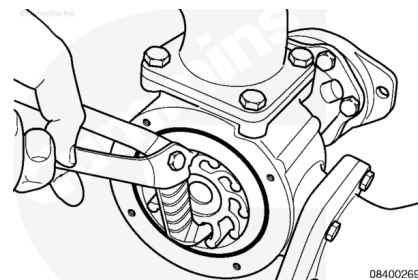
Remove the cover plate o-ring from the sea water pump housing.



⚠ CAUTION ⚠

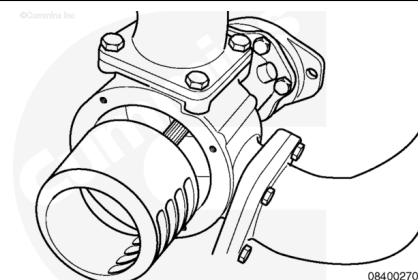
Do not pry against the cam liner to remove the impeller as this can cause damage to the liner. Be sure to note the direction of impeller fins for proper reinstallation.

Grasp the hub of the impeller with pliers and remove it from the impeller bore.

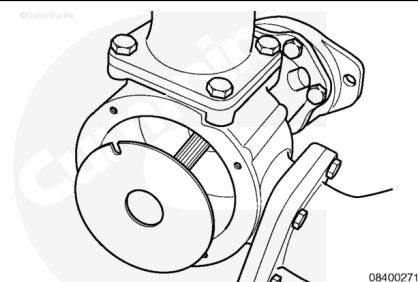


Note : Care should be taken **not** to drop or damage the cam liner.

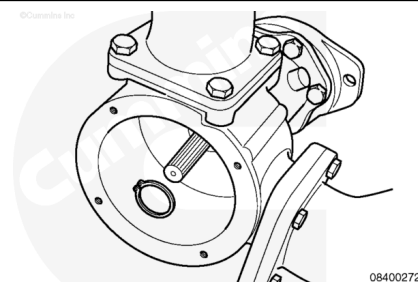
Remove the cam liner from the pump body.



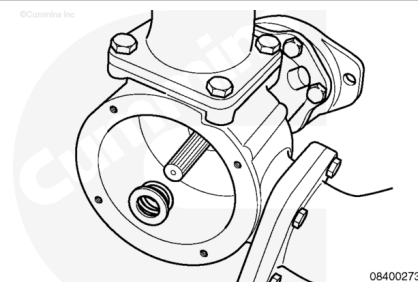
Remove the inner wear plate.



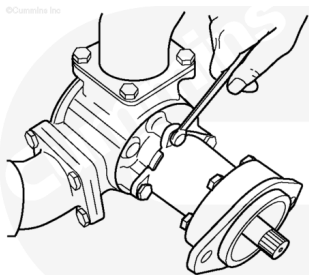
Remove the mechanical seal-retaining snap ring from the shaft.



Remove the spring and plate, and discard the parts.



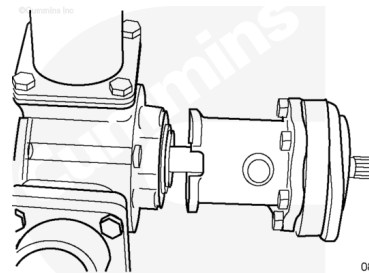
Remove the four bolts with washers mounting the bearing housing to the impeller housing use a 13-mm wrench.



08400274

Note : An o-ring will remain on the bearing housing side of the shaft when the two housings are separated. This o-ring should be inspected before reuse.

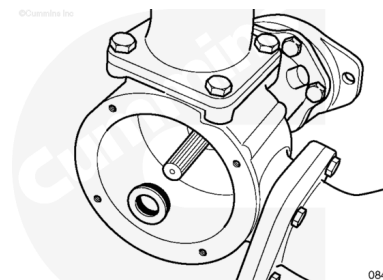
Separate the impeller housing from the bearing housing. This is performed to remove the mechanical seal from the impeller shaft inside the impeller housing.



08400275

Once the bearing housing has been pulled back, as in the previous step, the carbon sealing ring can easily be removed from its recessed area in the impeller housing.

Note : Remove and discard the carbon sealing ring.

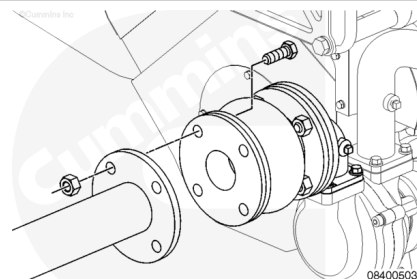


08400276

Gilkes

Remove the four capscrews and nuts from the inlet connection and separate the sea water piping to the inlet connection.

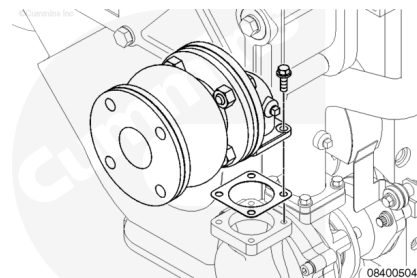
Discard the gasket.



08400503

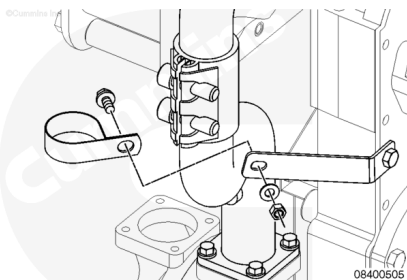
Remove the four capscrews from the inlet elbow that connects to the sea water pump and remove the inlet connection and elbow.

Discard the gasket.

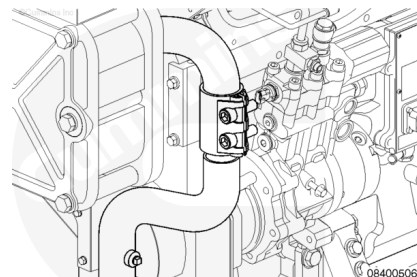


08400504

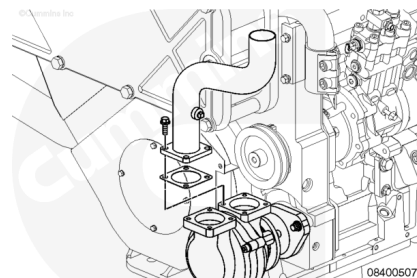
Remove the pipe clamp supporting the water transfer tube.



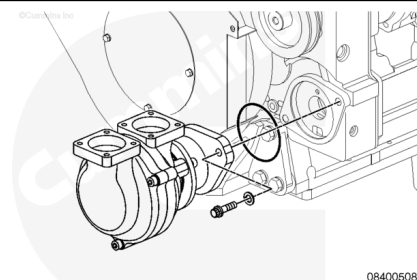
Loosen the two capscrews securing the flexible coupling and slide it up onto the upper portion of the transfer tube.



Remove the four capscrews, lower transfer tube, and gasket. Discard the gasket.



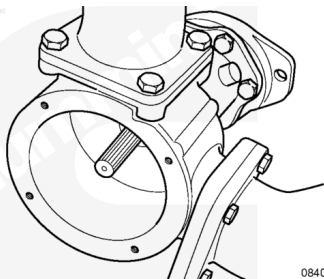
Remove the two mounting capscrews, sea water pump and o-ring. Discard the o-ring.



Inspect for Reuse

Jabsco®

Inspect sea water pump housing, impeller shaft, and sealing surfaces for damage or wear. Clean reusable parts.

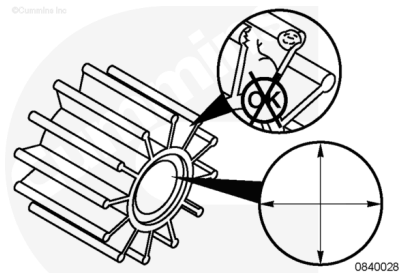


08400277

If the original impeller is to be reused, carefully inspect for damage such as rips, tears, chunks of material missing, or wear on the edges of the blades. Carefully inspect the cam liner for damage.

Replace parts as necessary.

©Cummins Inc.



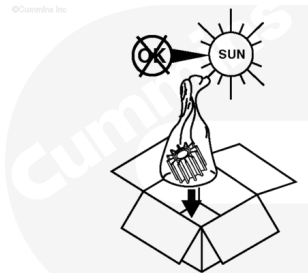
08400280

Seasonal Storage

If the impeller is in good shape and is to be used again, follow these storage tips:

- The impeller **must** be lubricated with glycerin or a non-petroleum-based lubricant such as silicone spray
- Store the impeller for the winter season in a polyethylene bag, preferably in a dark location
- Avoid exposure of the impeller to sunlight or any other source of ultraviolet light
- If the impeller is damaged or is **not** to be installed, do **not** install the new impeller until the vessel is ready to be used (the following season).

©Cummins Inc.



08400281

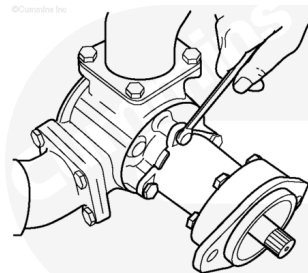
Install

Jabsco®

Note : Make sure the o-ring slinger is installed on the shaft before installing the bearing housing.

Install the sea water pump bearing housing to impeller housing with four bolts and lock washers, use a 13-mm wrench.

©Cummins Inc.



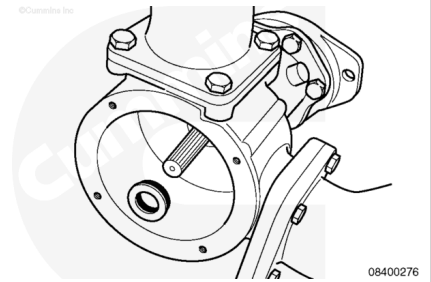
08400274

⚠ CAUTION ⚠

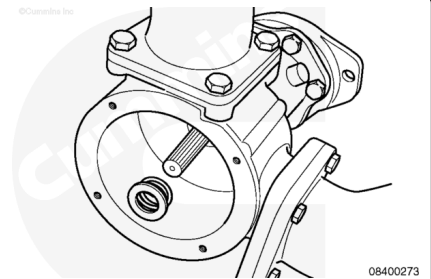
Do not touch the carbon face of the mechanical seal, or leaks and damage can occur.

Install the new mechanical seal onto the sea water pump shaft. Seat the seal into the housing recess.

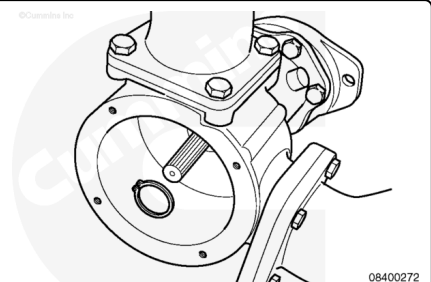
Note : Never reuse a mechanical seal. Always install a new seal, spring, and plate.



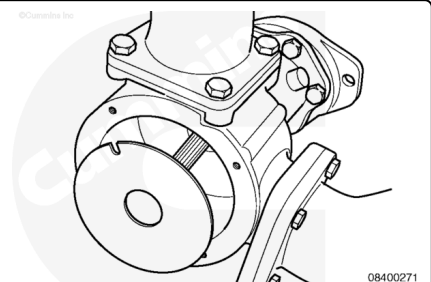
Install the spring and plate.



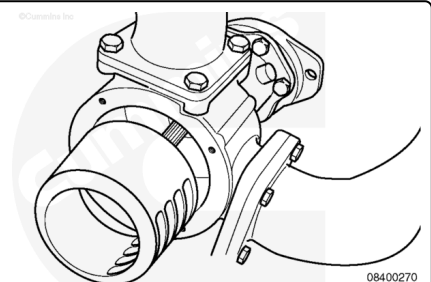
Install the seal-retaining ring onto the shaft.



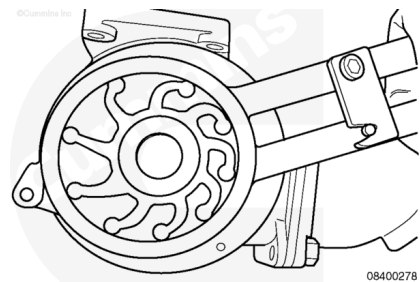
Install the inner wear plate and align the dowel pin in the housing.



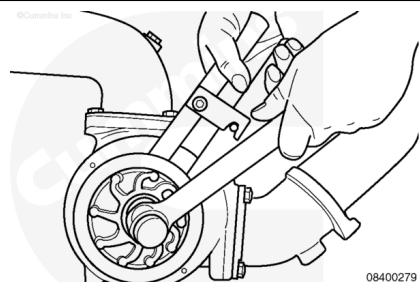
Install the cam liner into the sea water pump impeller housing, aligning the hole in the liner with the dowel in the housing.



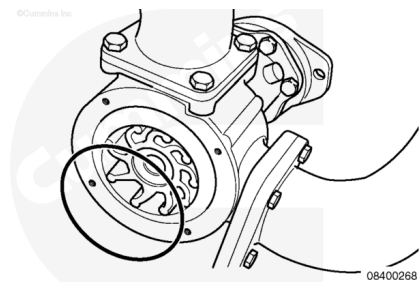
Roll the impeller into an oil filter wrench with the fins facing the opposite direction of the sea water pump rotation.



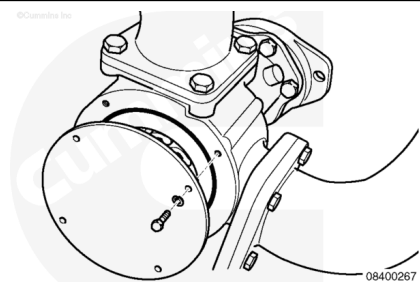
Install the impeller onto the cam liner and onto the splined shaft.



Coat the o-ring with engine oil. Install the cover plate o-ring into the groove in the housing.



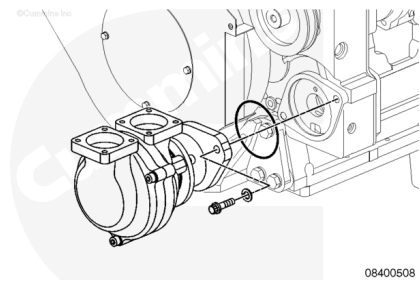
Install the sea water pump front cover with four bolts and lock washers use a 10-mm wrench.



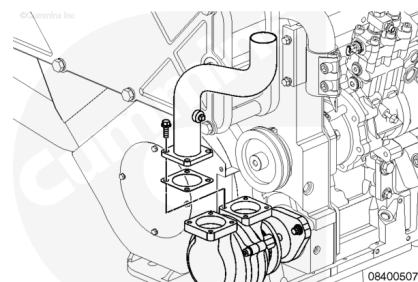
Gilkes

Install the sea water pump, new o-ring, and mounting capscrews.

Mounting Capscrews	80 n.m	[60 ft-lb]
--------------------	--------	------------

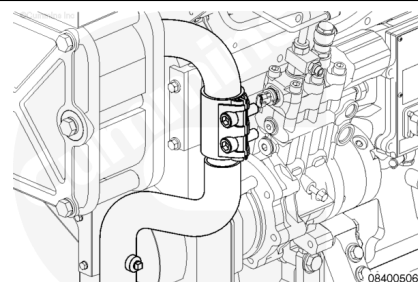


Install the lower transfer tube, new gasket, and four capscrews.



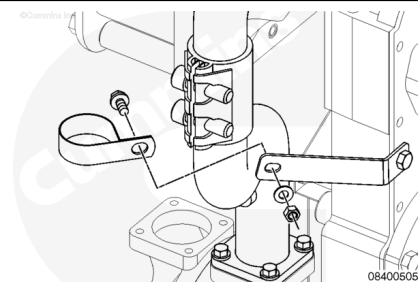
Lower Transfer Tubes Capscrews	45 n.m	[33 ft-lb]
--------------------------------	--------	------------

Slide the flexible coupling down onto the lower portion of the transfer tube covering the gap between the upper and lower transfer tubes. Tighten the two coupling capscrews.



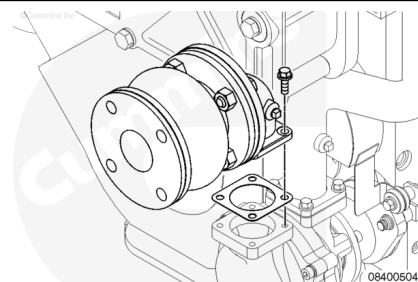
Coupling Capscrews	45 n.m	[33 ft-lb]
--------------------	--------	------------

Install the pipe clamp supporting the water transfer tube.



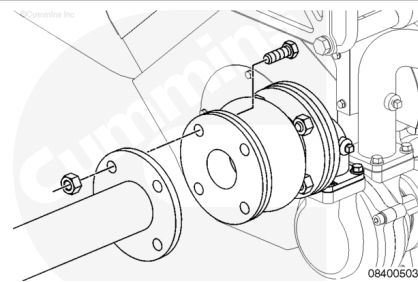
Pipe Clamp Supporting	40 n.m	[30 ft-lb]
-----------------------	--------	------------

Install the inlet connection and elbow, gasket, and the four capscrews to the sea water pump.



Inlet Connection Capscrews	45 n.m	[33 ft-lb]
----------------------------	--------	------------

Install the sea water piping, new gasket, and the four capscrews and nuts to the inlet connection.

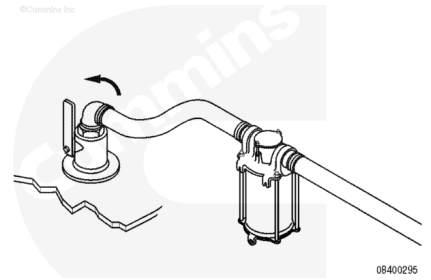


Sea Water Piping Capscrews	80 n.m	[60 ft-lb]
----------------------------	--------	------------

Finishing Steps

Open the sea water inlet valve. Check for leaks.

Operate the engine and check for leaks.



Last Modified: 03-Aug-2006
